Sebastian Horta

New York | (516) 974-2139 | sh4506@columbia.edu | linkedin.com/in/sebastianhorta03 | github.com/SebasHorta | Portfolio

Education

Columbia University (SEAS) New York, NY
B.S. Computer Science, Minor: Electrical Engineering *Aug 2023 – May 2026*

- Coursework: Computer Networks, NLP, Adv. C++ Systems Programming, Competitive Programming, Artificial Intelligence, Data Structures & Algorithms, Systems Programming, CS Theory, UI Design, Computer Systems
- Resident Adviser: Organized 10+ community events for 700+ residents, leveraging cross-functional teamwork

Experience

Machine Learning Engineer — Columbia University Robotics Lab *2025*
New York, NY

- Built **ML** pipeline in **Python** to detect gait events (heel strikes/toe offs) from VR motion-tracker data
- Engineered unsupervised **KMeans** clustering to classify VR trackers by body segment and foot side
- Developed **cross-correlation-based synchronization** between VR tracker streams and MAT ground truth
- Trained **Random Forest** classifier, correctly detecting ∼**85%** of gait events from VR tracker data

Full Stack Software Developer — Freelance *2025 – 2026*
Remote

Web Archiver Tool | **React + FastAPI** 2025

- Built offline web archiver with structured storage and versioning for **research** and offline site analysis
- Developed **recursive crawler** using **requests + BeautifulSoup** for HTML and asset capture
- Implemented URL rewriting and error handling for irregular link structures
- Created **React** dashboard with real-time progress tracking and iframe previews

Client Business Website (Live Site) | **Next.js + TypeScript + Tailwind** 2026

Projects

Graph-Scoped Context Server for LLM Coding Agents | **Python + MCP** 2026

- Built an **MCP server** that scopes AI coding-agent context using dependency graphs and git-diff filtering
- Built a **benchmark harness** with 30+ faults to compare graph-based and baseline retrieval
- Implemented **dependency-graph traversal**, automated bug injection, and live MCP tool integration
- Designed an evaluation pipeline measuring **token efficiency** and fault-localization accuracy

Allowance | **Python** 2026

- Designed a **P2P blockchain** enforcing AI agent spending limits via **ECDSA**-signed transactions and **PoW**
- Implemented **Merkle tree** transaction integrity and fork resolution to maintain consensus across 4+ nodes
- Built a peer discovery tracker with heartbeat monitoring, persistent **TCP** connections, and thread-safe state
- Validated correctness with **106 tests** covering transaction signing, overspend rejection, and tamper resistance

Skills

Languages & Frameworks: Python, Java, C/C++, JavaScript/TypeScript, HTML/CSS, React, Next.js, Tailwind
Backend & AI/ML: FastAPI, Flask, Node.js, REST APIs, SQL, scikit-learn, pytest, OpenCV, LLMs
Tools: Git, Unix/Linux, CI/CD, Figma

Leadership & Activities

SHPE at Columbia – Committee Board Member – Planned cultural and career events, resulting in increased presence on campus

IEEE, ColorStack, CU ADI, CU BJJ Club, Sabor Dance Club – Member – Engaged in club events, community building, and collaborative projects

Collegiate Science and Technology Entry Program (CSTEP) – Member – Participated in program initiatives and STEM outreach activities

Awards

Columbia Spirit Award (2026) • Anders Scholarship (2024) • Columbia University Nathaniel Arbiter Scholarship (2024)
FSC President's List (2022–23) • NYS Scholarship for Academic Excellence (2022)